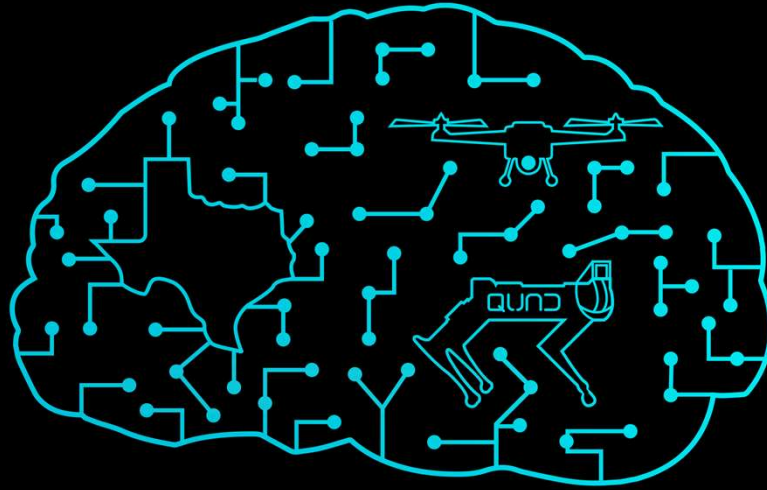




Preparing Students for the Future World of AI



ALEX SEGUIN

EdTech Innovation Manager, Part-time AI/ML Technology Solutions

US Air Force Veteran, Engineer, MBA, AI Education

Product Portfolio Management Process Overview

01

Product Management Overview

What is Product Management?

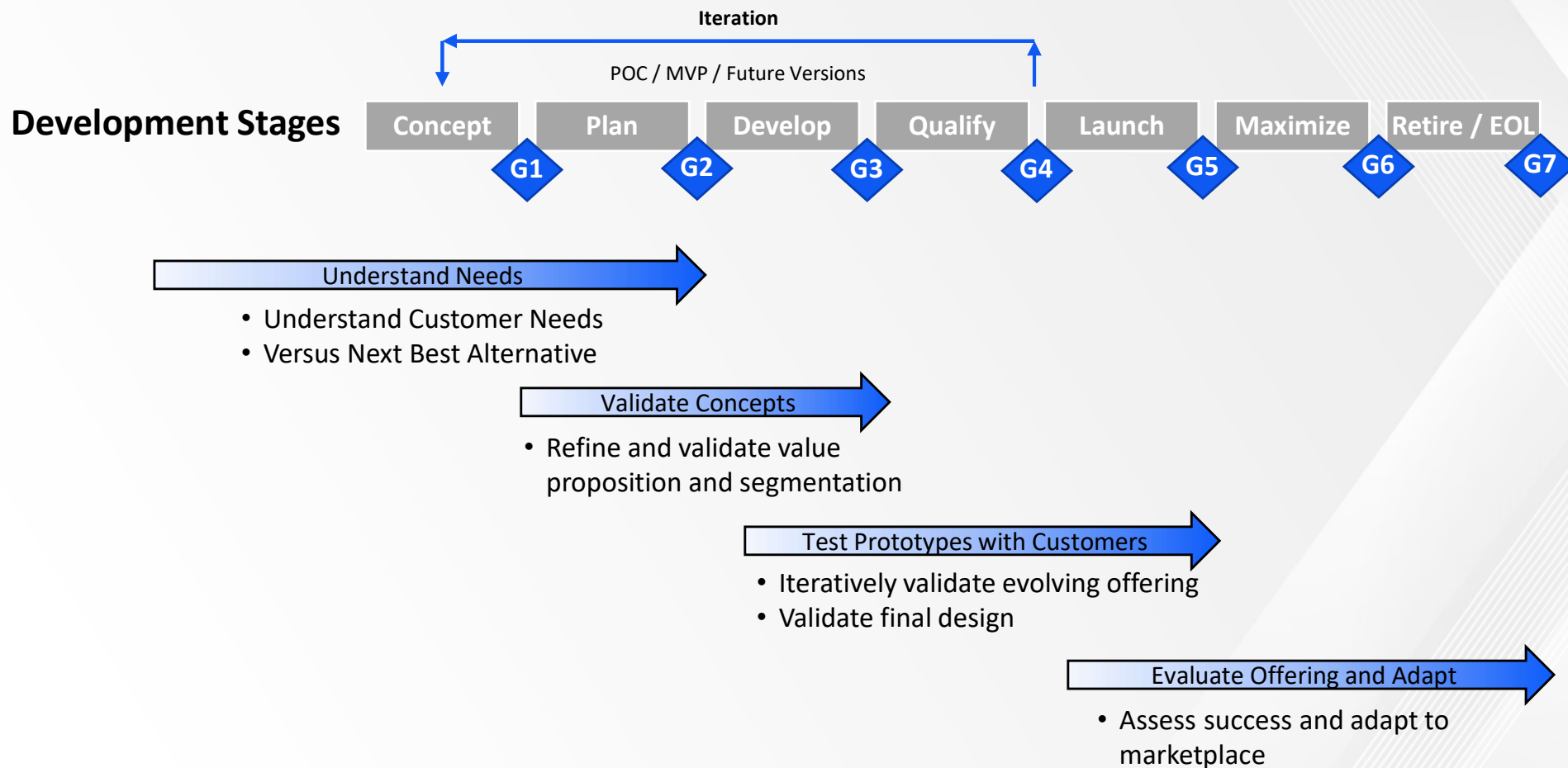
What is Product Management?

A Product Manager makes outward, market facing decisions.



Product Management is..... a business function that acts as the connective tissues between corporate stakeholders to ensure products are brought to market in an efficient and profitable manner.

Robust Stage-Gate: Collaborative Input Improves NPD Success



Product Management Gate *Deliverables*

CROSS FUNCTIONAL INPUT

Product Management

FEEDBACK + DOCUMENTATION

Market Needs

Describes Customer Needs & Problems

Business Case

Evaluates Opportunity

Product Description

Describes What to Build

Development Roadmap

Determines Long-term Goals and Strategy

Market Strategy

Determines How to Take the Product to Market

EXECUTION

Stage / Gate	Description
Concept	Executive Summary
	Financial – Return
	Customer CONOPS
	Customer Problem
	Product Summary
Plan	Customer Workflow
	Competition + Value
	User Stories - Highlights
	Financials by Year
	Prelim Feedback
Develop	Go-To- Market
	Risk Highlights
Qualify	Resourcing
	BETA Feedback
Launch + Maximize + Retire	Market Performance
	Post-Launch Dev
	End-of-Life Strategy

REVIEWS & APPROVALS

02

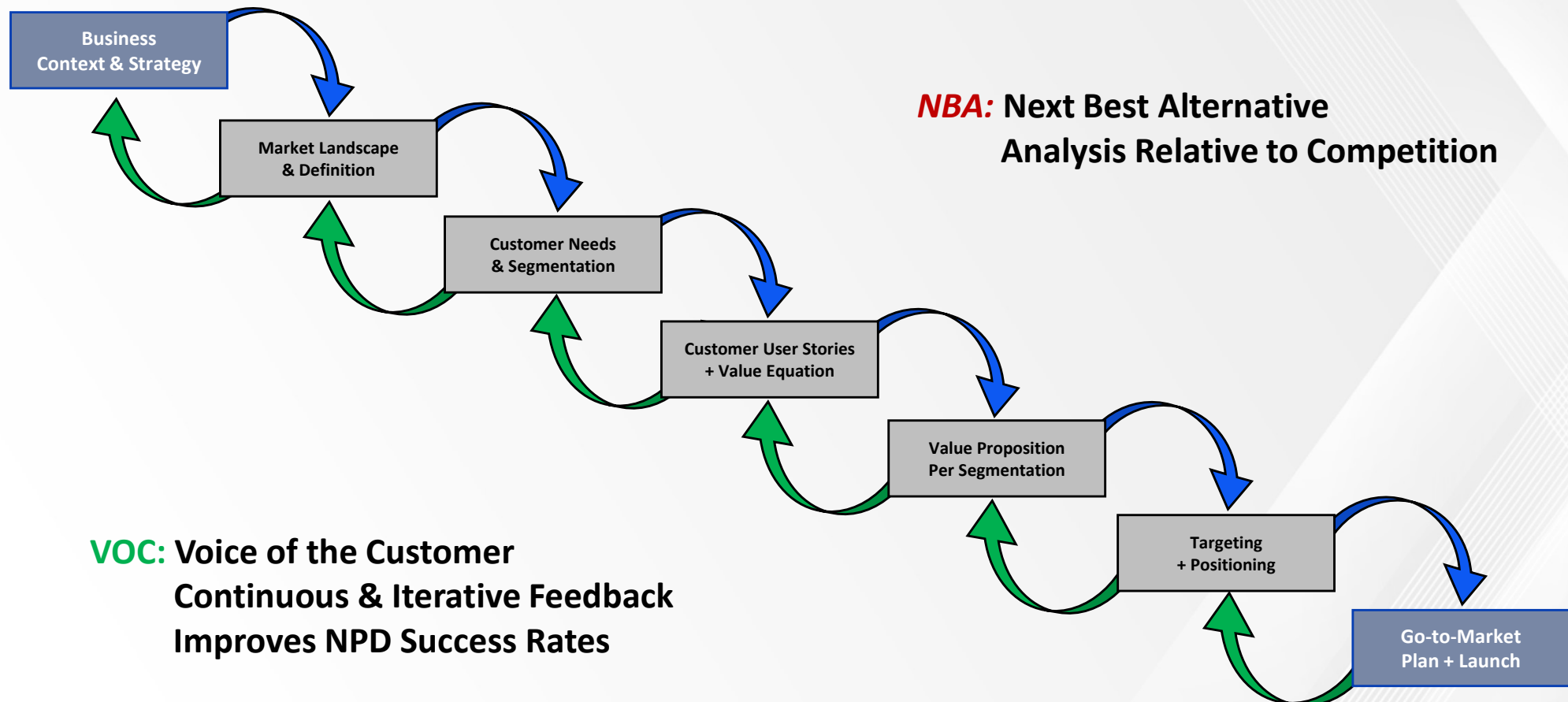
Portfolio Management Overview

Portfolio Objectives, New Product Strategy, Portfolio Review By Stage, X-Functional Virtual Teams, Reviews + Approvals, Portfolio Status

Product Portfolio Management Objectives

To inspire and sustain an innovative culture that regularly prioritizes new AI and technological capabilities that add additional value to our customers above and beyond our competitors.

New Product Strategy: Addresses Risks vs Competitive NBA



Product Portfolio Review *Overview By Stage*



Stage / Gate	Slide #	Description	Summary	Preparation + Alignment
Concept	S1 - S2	Executive Summary	• 2 Slide Overview	• Must Have - Market Needs Documentation - Business Case Documentation - Alignment meeting: S1 - S6
	S3	Financial – Return	• Profile: NPV + IRR	
	S4	Customer CONOPS - Applications	• Mission Needs	
	S5	Customer Problem	• Statement-Outcomes	
	S6	Product Summary	• Customer Segment 1	
Plan	S7	Customer Workflow	• By Persona	• Must Have - Product Description Documentation - Alignment meeting: S6 – S11
	S8	Competition + Value	• Value Prop v NBA	
	S9	User Stories - Highlights	• By Persona	
	S10	Financials by Year	• Cost + Revenues	
	S11	Prelim Feedback	• Validation + Plans	
Develop	S12	Go-To- Market	• Strategy Alignment	• Must Have - Roadmap Documentation - Alignment meeting: S12-S13 • Recommended: - Manufacturing Strategy Documentation - Supply Chain Risk Assessment
	S13	Risk Highlights	• Tech + Commercial	
Qualify	S14	Resourcing	• Functional Alignment	• Must Have - Marketing Strategy Documentation
	S15	BETA Feedback	• Pre-Launch VOCs	
Launch + Maximize + Retire	S16	Market Performance	• Say / Do Metrics	• Recommended: - Performance Tracking w/ Finance + BD - Performance Tracking w/ Mfg/Ops
	S17	Post-Launch Dev	• VOC + Priorities	
	S18	End-of-Life Strategy	• Org Alignment	

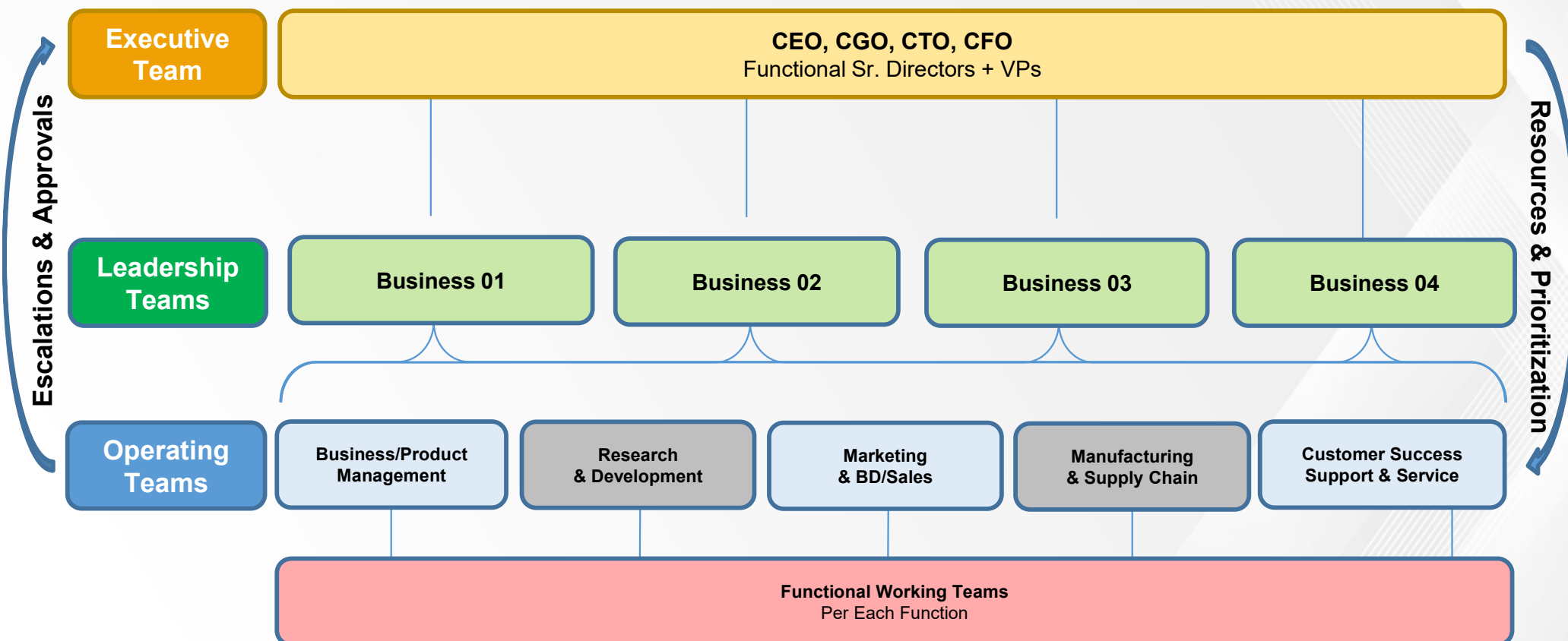
Functional Contributions DACI + Approvals

Team	Product Portfolio	Owner	Mark Holden	DACI + Approvals by Gate								● Approver ○ Reviewer								
Functional Approvals				PRB	Concept	PD	Plan	PRN	Develop	NS	Qualify		Launch	Maximize	Iterate/Retire					
PRB: Product Review Board		Target Dates -->			MM/YY		MM/YY		MM/YY		MM/YY			MM/YY		MM/YY				
Business Management					D		●		D		●		D		●		D		●	Mark Holden
Finance / FP&A					I		○		A		●		A		●		C		○	Alex Higgs
R&D – Hardware R&D - AI / ML Architecture R&D/ IT Sec-Eng / Cyber Security					C		●		A		●		A		●		I		●	Christian Wester Pete Lommel Mike Fourie
Testing Operations Team					I		○		I		●		C		●		A		●	Mike Fourie
UXD - User Experience / Design					A		○		A		○		C		●		A		○	Irene Tena
Manufacturing / Supply Chain					I		○		I		○		C		○		A		●	Manas Menon
Marketing + Asset Management					I		○		I		●		I		○		A		●	Nicole Rizzi
Business Development / Sales					I		●		C		●		A		○		A		●	Chris Brinkley
Customer Support / Service					I		○		A		●		I		○		A		○	Justin Maphet
Human Resources					I		○		I		○		I		○		I		○	Scott Chaplain
Legal / Trade Compliance					C		○		I		○		C		●		A		○	Jamie Brown

D - Driver: Completes task, organizes others to Accountable
A - Accountable: Signs off that the task is complete
C - Contributor: This party's input/resources are requested at some point during the process
I - Informed: This party is informed of the outcome

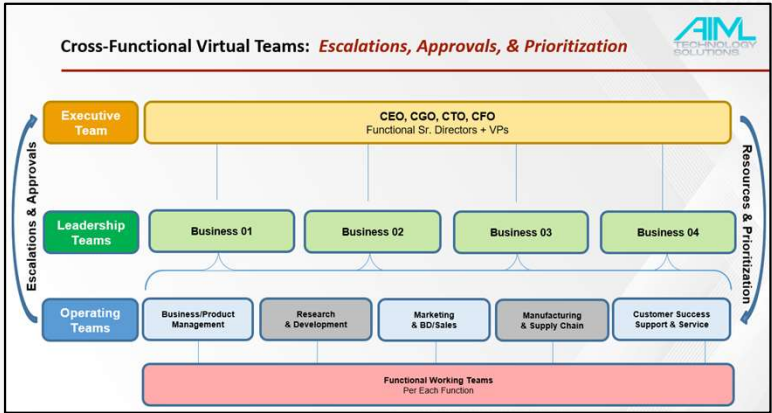
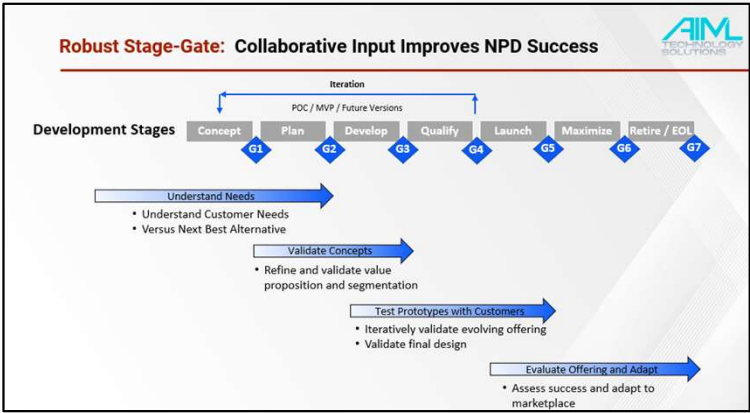
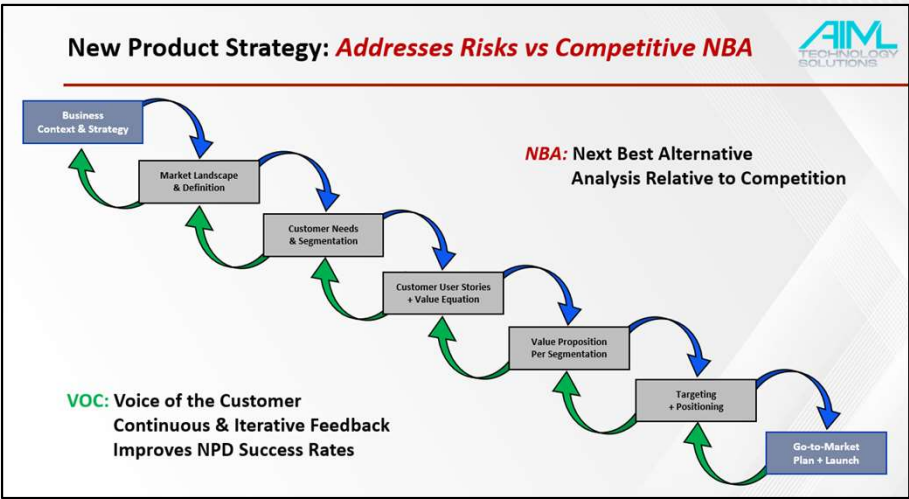
Pre-Concept - Ideation: At least 2x / Year
 Legal - Launch: FAA, ITAR, License for DFM

Cross-Functional Virtual Teams: *Escalations, Approvals, & Prioritization*





Product Management Summary



03

Businesss Case Review

Executive Summary + Detailed Deliverables for Review

PROJECT NAME

COGS\$ -- kMSRP\$ -- kMgn %%

Image

Functional Lead

Name

COGS

To-Date

2022

Life

Product MgrF. LastI-RAD-- -- --

Project EngF. LastC-RAD-- -- --

BD/Sales LeadF. LastCap-Ex-- -- --

Financials

1st Rev DateQ# - YY

3-Year NPV\$ -- M

IRR%

Research/Demo

Feature/Payload

NextGen/Refresh

New Product

Sustain

Project Overview

Project Summary

Project A is a X, Y, Z designed to X, Y, Z from a A or B or C platform.

In YYYY, Project A seeks to enter Full Rate Production, on the X + Y model variants.

Market Opportunity

Franchise Pursuits

Pursuit

Award

Full value

Prog 1YYYY\$ - M

Prog 2YYYY\$ - M

Prog 3YYYY\$ - M

CRM: Sales Force

Date

Award

Full value

Q# - YYYYYY\$ - M

Q# - YYYYYY\$ - M

Q# - YYYYYY\$ - M

Q# - YYYYYY\$ - M

Comments

X

X

X

X

Portfolio Positioning

Price + Performance

600,000

500,000

400,000

300,000

200,000

100,000

0

PRD NBA 2

SHIELD AI

Product NBA 1

Low

Med

High

Product Strategy / Value Proposition

Product Strategy

X

Y

Z

Value Proposition

X

Y

X

X

X

X

X

X

X

X

X

X

X

X

X

X

Dependencies: Intra-BU/LOB + Customer

Shield AI

Project

Project #

Prog 1YYYY

Prog 2YYYY

Prog 3YYYY

DoD - Customer

Project

Project #

Prog 1YYYY

Prog 2YYYY

Prog 3YYYY

Roadmap + Schedule

LINK TO TEMPLATE

Shield AI - MVP vs NBA: Competitor B Product

100.0

30.0

20.0

10.0

8.0

3.0

2.0

12.0

11.0

25.0

125.0

Value Creation

Value Capture

\$ 50 k

50%

Project Description

Project Summary

- Project A is a X, Y, Z designed to X, Y, Z from a A or B or C platform.
- In YYYY, Project A seeks to enter Full Rate Production, on the X + Y model variants.

Image

Information

Segment

- X

Functional Lead	Name
Product Mgr	F. Last
Project Eng	F. Last
BD/Sales Lead	F. Last

Product Variant 1

COGS	MSRP	Mgn %
\$ -- k	\$ -- k	%

Product Variant 2

COGS	MSRP	Mgn %
\$ -- k	\$ -- k	%

Comments

- X
- Y
- Z

Objectives

2022

- X
- Y
- Z

2023 + Beyond

- X
- Y
- Z

Related Activities

- X
- Y
- Z

Dependencies

Shield AI: Internal I-RAD

- X
- Y
- Z

Shield AI: Internal C-RAD

- X
- Y
- Z

External: Customer

- X
- Y
- Z

Critical Issues & Risks

Risk	Issue	Counter Measure / Mitigation	Owner
C1	X	X	F. Last
C2	X	X	F. Last
M1	X	X	F. Last
M2	X	X	F. Last
M3	X	X	F. Last
M4	X	X	F. Last
M5	X	X	F. Last

Project Cost Actuals vs Forecast: k\$

R&D Actuals

R&D Forecast

R&D Spend	2022
Forecast	\$ -- k
Actuals	\$ -- k

Year	R&D Actuals	R&D Forecast
2021	1,000	1,000
Q1-22		1,250
Q2-22		1,500
Q3-22		1,750
Q4-22		2,000
2023		2,250
2024		2,500
2025		2,950

Roadmap + Schedule

Research/Demo

New Product

Feature/Payload

NextGen/Refresh

Sustain

REQUIRED: **CONCEPT** and above

Financial Return – Profile: *Project Name*

Business: X

Project #: XYZ

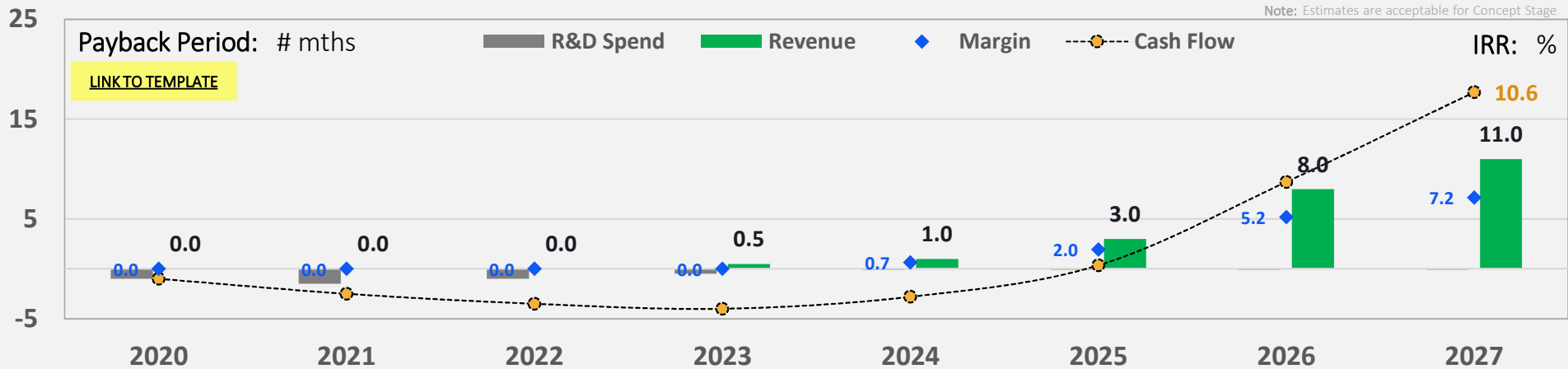
Slide: 3

3-Year NPV: X.X

Dev Stage: XYZ

Date 1st Revenue: MM/YYYY

Financial Chart, \$M



R&D Spend & Revenues

R&D	Past Total	2022	3 Year	5 Year	Incremental	2022	3 Year	5 Year	3 Year NPV
Spend	\$ -- M	\$ -- M	\$ -- M	\$ -- M	Full Revenues	\$ -- M	\$ -- M	\$ -- M	\$ -- M
Financial Comments: - X - Y					Full Margin	\$ -- M	\$ -- M	\$ -- M	
					Prob Weighted Revenue	\$ -- M	\$ -- M	\$ -- M	\$ -- M
					Prob Weighted Margin	\$ -- M	\$ -- M	\$ -- M	

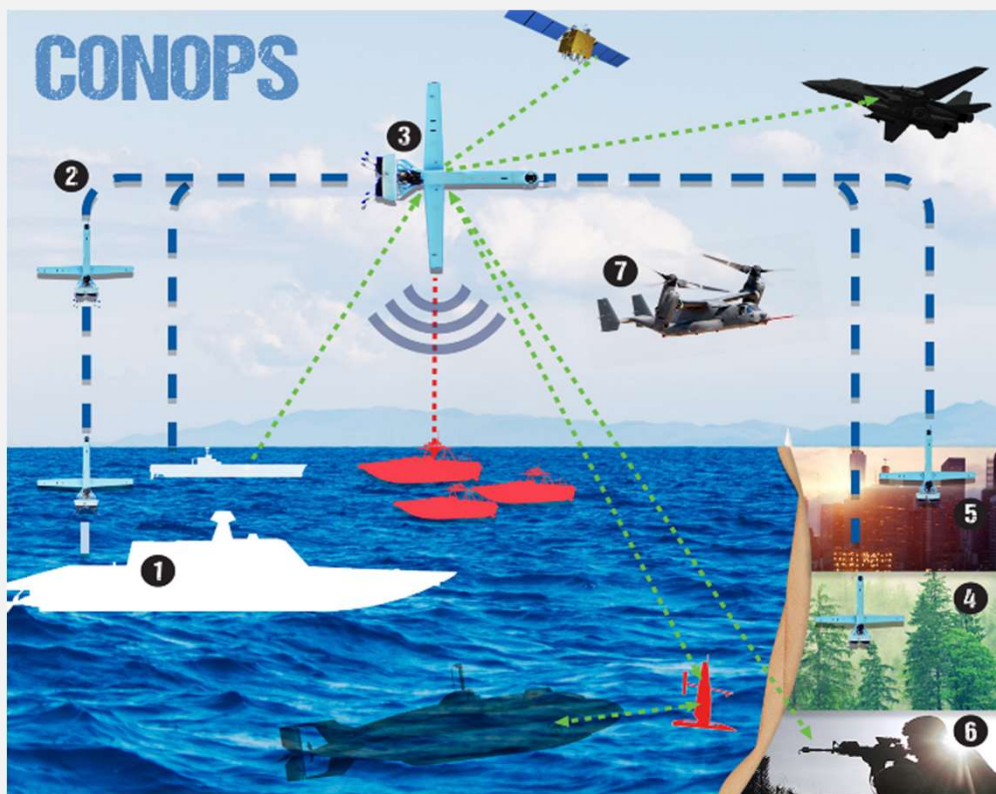
Technical Risk: LOW Commercial Risk: MED

Customer - CONOPS: *Project Name*

Business: X Project #: XYZ Slide: 4 (Source: Market Needs)



CONOPS



1

Ability to take-off from / delivery to large, medium, and small watercraft

2

Vertical take-off in up to 30 kt winds

3

Communicate with and receive guidance from a range of Joint sources

4

Cargo delivery to constrained landing zones and austere locations

5

Delivery to rooftops, courtyards, and other locations to support urban warfare or HADR

6

In extremis delivery of essential equipment or supplies to tactical Warfighters

7

Entire compact V-BAT system easily be delivered to a ship or other point of need

Future Capabilities

GPS-denied navigation, landing, and collaborative behaviors

Diagrams: <https://assets.shield.ai/shield-ai>

Example: Modify As Needed

Customer Problem Statement: *Project Name*

Business: X

Project #: XYZ

Slide: 5 (Source: Market Needs)

1	Problem Statement	- Warfighters involved in a dynamic time-sensitive targeting situation need faster intelligence, accurate target acquisition, automation of manual work to enable quicker and safer decision making
2	Customer Outcomes	- Accuracy in locking onto target (HVI, vehicle, etc) - Faster and accurate risk calculations - Better visual situational awareness of threats and risks
3	Customer Why's	- No CIVCAS - eliminate risk to collateral damage - Automation of multiple sources of data to reduce human error - Need higher confidence to decide on strike
4	Problems with the Status Quo	- Potential for human error is high (manual calculations, maintaining lock on moving target) - Current software can be cumbersome and lagging, not built to automate - Not enough people resources to support multiple ops happening simultaneously

Reference / Example Only

Product Summary: *Project Name*

Business: X

Project #: XYZ

Slide: 6 (Source: Business Case)



Product Description

Single Sentence Overview



Problem Statement

X

CONOPS + APPLICATIONS

X

Competition + Value Prop: *Project Name*

Business: X

Project #: XYZ

Slide: 8a (Source: Business Case)



Competitive Differentiators

Competitors

- Company 1, Company 2, Company 3

Key Customer Benefits

- X
- Y
- Z

Key Technical Features

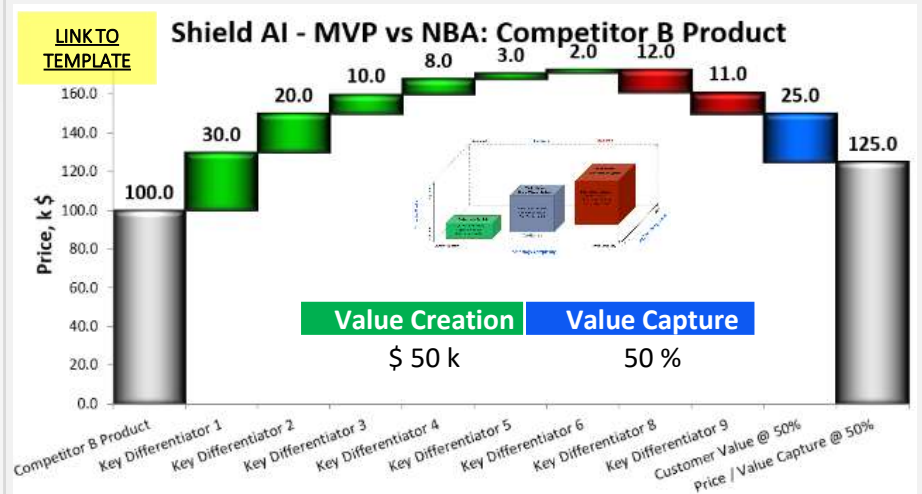
- X
- Y
- Z

May Also Use Performance Curve:
See Executive Summary

LINK TO TEMPLATE	Differentiators	Customer Importance	Max Value Creation	Shield AI MVP1 Offering	Primary NBAs Product 1	Product 2	Secondary NBAs Product 3
			\$150,000				
Customer Value 1							
Benefit 1							
- Differentiator 1			\$30,000	●	○	○	○
- Differentiator 2			\$20,000	●	○	○	○
- Differentiator 3			\$10,000	●	○	○	○
- Differentiator 4			\$5,000	○	●	○	○
- Differentiator 5							
Benefit 2							
- Differentiator 6			\$20,000	○	●	○	○
- Differentiator 7			\$10,000	○	●	○	○
- Differentiator 8			\$5,000	○	●	○	○
- Differentiator 9			\$3,000	○	●	○	○



Primary Customer Value Prop



Technical Differentiators	Customer Importance	Relative Performance	Differentiator Details Shield AI vs Company 1	Quantified Value Δ \$
Key Differentiator 1		▲	SOA vs C1	\$30,000
Key Differentiator 2		▲	SOA vs C1	\$20,000
Key Differentiator 3		▲	SOA vs C1	\$10,000
Key Differentiator 4		▲	SOA vs C1	\$8,000
Key Differentiator 5		▲	SOA vs C1	\$3,000
Key Differentiator 6		▲	SOA vs C1	\$2,000
Key Differentiator 7		▲	SOA vs C1	\$0
Key Differentiator 8		▼	SOA vs C1	(\$12,000)
Key Differentiator 9		▼	SOA vs C1	(\$11,000)

Value Proposition + Differentiators: **Sub-T Phase III Example**

Business: X

Project #: XYZ

Slide: 8b (Source: Business Case)



Competition + Benefits

Key Customer Benefits: **Sub-T Phase III**

- Accelerated mapping of underground facility
- Enhanced operational reach and communications
- Real time blue force tracking
- Advanced air quality monitors to increase survivability

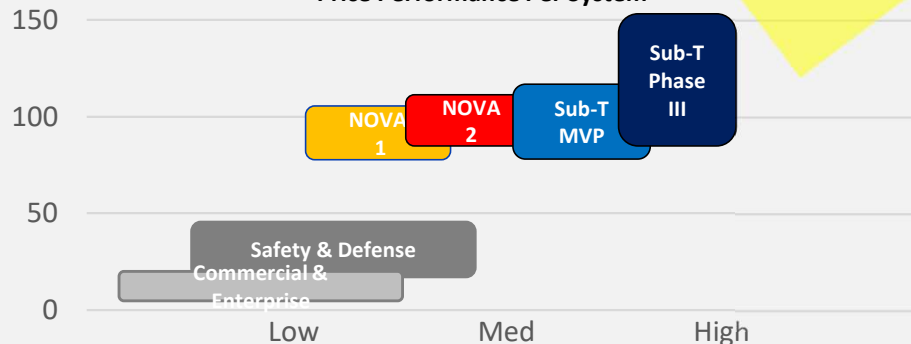
Key Technical Features: **Sub-T Phase III**

- Multi aircraft teaming with integrated network and advanced communications relay
- Modular payloads – kinetic, air monitoring sensors (O2, CO2, Methane, and Cyanide)
- Autonomous mapping, flight and auto return leveraging onboard GPU and sensor suite
 - Thermal imaging 640, stereo depth camera, 1080p video

Competitors

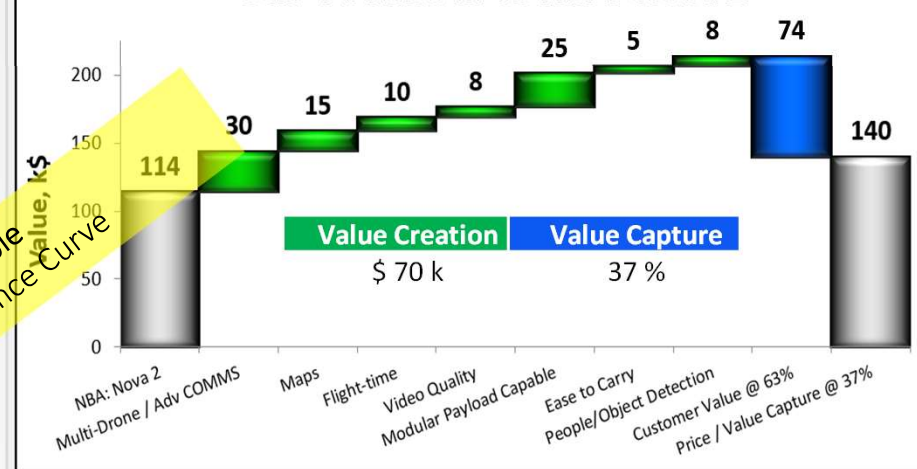
- **Commercial/Enterprise:** DJI – Mavic Series, Skydio X2
- **Safety/Defense:** Brinc - Lemur, Skydio - X2D

Price Performance Per System



Value Proposition + Differentiators

Sub-T Phase III vs NBA: Nova 2



Technical Differentiators	Customer Importance	Relative Performance	Differentiator Details Sub-T Phase III vs Nova 2
Multi-Drone / Adv COMMS	High	High	ATAK, BFT, COMMS Relay, Upgraded COMMS vs 1x Nova 2
Maps	High	High	User selected 2D/3D with multi level option vs 3D Stereodepth
Flight-time	High	High	20 min vs 15 min
Video Quality	High	High	1080p at 30 fps vs 720p at 30 fps
Enhanced Low Light Ops	High	High	Thermal Boson 640 + NFIR vs Thermal Boson 640 + NFIR
Modular Payload Capable	High	High	4x Gases Air monitoring + Kinetic vs None
Ease to Carry	High	High	Ease of carry vs Nova 2
Obstacle Avoidance	High	High	Manual/Autonomous-Day/Night vs Manual/Autonomous-Day/Night
People/Object Detection	High	High	Auto Person/Object Detection vs Auto Person Detection
Indoor Ease of Use	High	High	< 10 min training vs < 10 min training



User Story - Highlights: *Project Name*

Business: X

Project #: XYZ

Slide: 9 (Source: PDD Deck)



Personas

Persona 1:

- Brief Description

Persona 2:

- Brief Description

Persona 3:

- Decision Maker(s)
- Brief Description



High Priority User Stories



Persona 1:

- As a **Persona 1**, I want < some goal > so that < some reason > .
- As a **Persona 1**, I want < some goal > so that < some reason > .
- As a **Persona 1**, I want < some goal > so that < some reason > .

Persona 2:

- As a **Persona 2**, I want < some goal > so that < some reason > .
- As a **Persona 2**, I want < some goal > so that < some reason > .

Persona 3:

- As a **Persona 3**, I want < some goal > so that < some reason > .
- As a **Persona 3**, I want < some goal > so that < some reason > .

Financials - By Year: **Project Name**

Business: X

Project #: XYZ

Slide: 10a - **PLAN**

R&D Spend

Year 1 -->	L - 3 Year	L - 2 Year	L - 1 Year	Launch	Year 2	Year 3	Year 4	Year 5
Total	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M
SAI Labor	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Contractor	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Materials	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Other/Sustain	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---

• 2022 R&D Spend Request: \$ -- M

• Technical Confidence: 10%, 25%, 50%, 68%, 95%, 99%

• Commercial Confidence: XX %



R&D Revenues

Do Nothing: Existing	L - 3 Year	L - 2 Year	L - 1 Year	Launch	Year 2	Year 3	Year 4	Year 5
# Systems	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---
Revenue	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %
New: 1st Product Rev	L - 3 Year	L - 2 Year	L - 1 Year	Launch	Year 2	Year 3	Year 4	Year 5
Revenue	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M
Margin	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M
Declining Rev: Existing	L - 3 Year	L - 2 Year	L - 1 Year	Launch	Year 2	Year 3	Year 4	Year 5
Revenue	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Combined: Incremental	L - 3 Year	L - 2 Year	L - 1 Year	Launch	Year 2	Year 3	Year 4	Year 5
Revenue	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %

Reference Links: Doc 1, Doc 2, Doc 3

Annual Forecast Required
During **PLAN** Stage

Financials - By Month: **Project Name**

Business: X

Project #: XYZ

Slide: 10b - **DEVELOP**

R&D Spend

Year 1 -->	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Year 2	Year 3	Year 4	Year 5
Total	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M	\$ -- M
SAI Labor	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Sub-Contractor	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Material	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Other Direct	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Sustain	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---

• 2022 R&D Spend Request: \$ -- M

• Technical Confidence: 10%, 25%, 50%, 68%, 95%, 99%

• Commercial Confidence: XX %



R&D Revenues

Do Nothing: Existing	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Year 2	Year 3	Year 4	Year 5
# Systems	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---	# ---
Revenue	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %
New: 1st Product Rev	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Year 2	Year 3	Year 4	Year 5
Revenue	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M
Margin	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M	\$ --M
Declining Rev: Existing	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Year 2	Year 3	Year 4	Year 5
Revenue	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Combined: Incremental	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Year 2	Year 3	Year 4	Year 5
Revenue	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Margin %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %	Y %

Reference Links: Doc 1, Doc 2, Doc 3

Monthly Forecast Required
During **DEVELOP** Stage

Preliminary Feedback: *Project Name*

Business: X

Project #: XYZ

Slide: 11



Early Validation - UXD

# Customers Contacted	Value Prop Differentiator	Voice of the Customer Learnings	Decision
# ---	XYZ	What did we learn?	Pivot
# ---	XYZ	What did we learn?	Pivot
# ---	XYZ	What did we learn?	Pursue
# ---	XYZ	What did we learn?	Pursue
# ---	XYZ	What did we learn?	Pass
# ---	XYZ	What did we learn?	Pass

Comments

- W
- X
- Y
- Z

Reference Links: Doc 1, Doc 2, Doc 3



Planned Experiments

Exp #	# Responses	Assumptions to Test	Success Criteria	Results
# --	##	If X, then Y . . .	Experiment is successful if X.	How did experiment perform vs our desired success criteria?
# --	##	If X, then Y . . .	Experiment is successful if X.	How did experiment perform vs our desired success criteria?
# --	##	If X, then Y . . .	Experiment is successful if X.	How did experiment perform vs our desired success criteria?
# --	##	If X, then Y . . .	Experiment is successful if X.	How did experiment perform vs our desired success criteria?
# --	##	If X, then Y . . .	Experiment is successful if X.	How did experiment perform vs our desired success criteria?
# --	##	If X, then Y . . .	Experiment is successful if X.	How did experiment perform vs our desired success criteria?

Comments

- W
- X
- Y
- Z

Go-to-Market Strategy: *Project Name*

Business: X

Project #: XYZ

Slide: 12

Launch Phases: Days



L - 90	L - 60	L - 30	L = Launch
Draft Launch Plan	Align & Execute Launch Plan	Execute Final Launch Deliverables	Launch & Optimize
1. Identify 1st Pass Marketing Assets and Launch Priority Level (L1-L4) with draft dates 2. Establish Preliminary Budget Range (plan for upside spend) 3. Host alignment meeting to identify internal & external marketing resources 4. Schedule follow-up meetings to monitor execution leading to L - 60	1. Identify 2nd Pass Marketing Assets with aligned dates 2. Establish Approved Budget Range (with upside spend) 3. Host alignment meeting to execute to Launch Plan 4. Schedule follow-up meetings to monitor execution leading to L - 30, identify Risks.	1. Identify Final Marketing Assets with aligned dates 2. Finalize Approved Budget Range (lower/raise) 3. Continue executing to Launch Plan, escalate as needed. 4. Meet as needed to monitor execution leading to Launch, monitor & report out Risks.	1. Identify Final Marketing Assets with aligned dates 2. Report out on Budget Actuals vs Plan. Discuss optimization. 3. Complete remaining Launch Plan deliverables. 4. Optimize marketing efforts to meet revenue projections, continue to report out Risks.

Reference Links: Market Strategy Documentation, Doc 2, Doc 3

Risk Summary: **Project Name**

Business: X

Project #: XYZ

Slide: 13



Technical Risk Highlights

Risk	Topic	Counter Measure
High	XYZ	XYZ
High	XYZ	XYZ
High	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Low	XYZ	XYZ
Low	XYZ	XYZ
Low	XYZ	XYZ
Low	XYZ	XYZ

Dependencies – Internal IRAD / CRAD Projects

- Product X, Project #
- Product Y, Project #
- Product Z, Project #

Reference Links: Doc 1, Doc 2, Doc 3



Commercial Risk Highlights

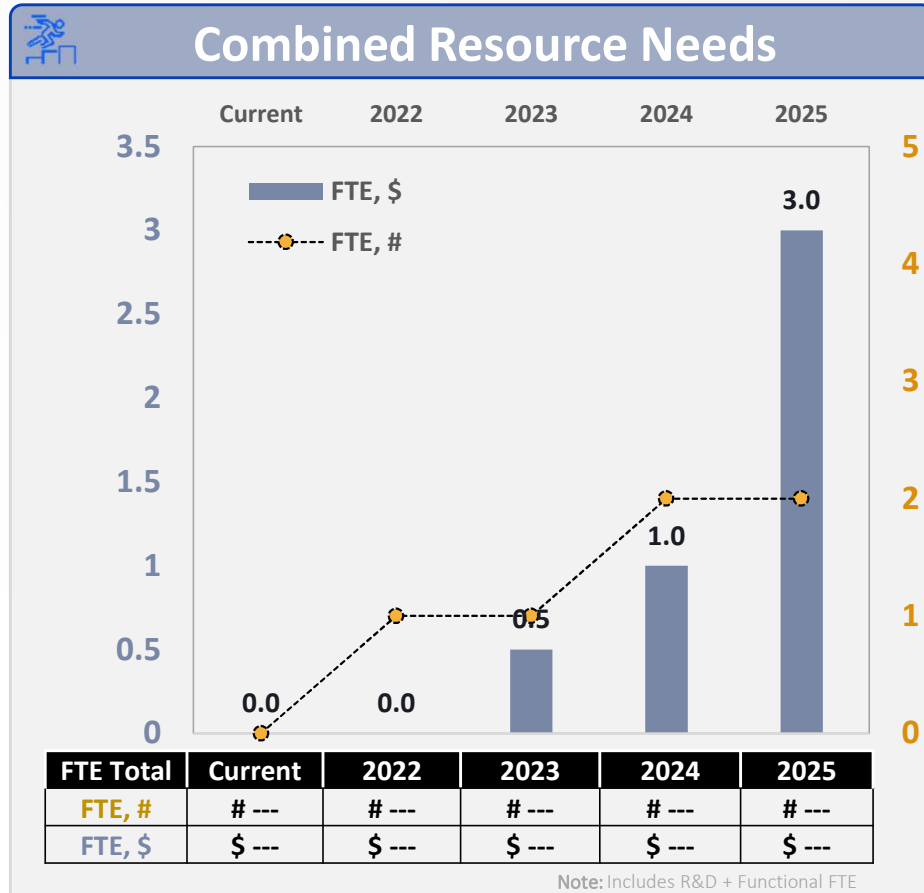
Risk	Topic	Counter Measure
High	XYZ	XYZ
High	XYZ	XYZ
High	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Med	XYZ	XYZ
Low	XYZ	XYZ
Low	XYZ	XYZ
Low	XYZ	XYZ
Low	XYZ	XYZ

Functional Resource Alignment: *By Project*

Business: X

Project #: XYZ

Slide: 14



Functional Alignment

FTE #	2021	2022	2023	2024	2025	2026	2027
R&D - Engineering	# ---	# ---	# ---	# ---	# ---	# ---	# ---
UX / ID	# ---	# ---	# ---	# ---	# ---	# ---	# ---
Mkt / Sales / BD	# ---	# ---	# ---	# ---	# ---	# ---	# ---
Mfg Ops / Supply	# ---	# ---	# ---	# ---	# ---	# ---	# ---
CS /Support / Service	# ---	# ---	# ---	# ---	# ---	# ---	# ---

- Technical + Manufacturing Resources
 - W
 - X
- Commercial Resources
 - W
 - X
- Supporting Infrastructure
 - W
 - X

FTE \$ Estimate	Current	2022	2023	2024	2025	2026	2027
Marketing / Sales	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Mfg Ops / Supply	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---
Support / Service	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---	\$ ---

BETA Feedback: *Project Name*

Business: X

Project #: XYZ

Slide: 15



Pre-Launch: BETA Test VOC

# Customers Contacted	Value Prop Differentiator	Voice of the Customer Learnings	Decision
# ---	XYZ	What did we learn?	Pivot
# ---	XYZ	What did we learn?	Pivot
# ---	XYZ	What did we learn?	Pursue
# ---	XYZ	What did we learn?	Pursue
# ---	XYZ	What did we learn?	Pass
# ---	XYZ	What did we learn?	Pass

Comments

- W
- X
- Y
- Z



Development Priorities

Priority	Feature Enhancement Summary	Method + Timing
High	Enhancement 1	Before Launch: MM/DD
High	Enhancement 2	Before Launch: MM/DD
Med	Enhancement 3	Field Update: MM/DD
Med	Enhancement 4	Field Update: MM/DD
Low	Enhancement 5	Other Update: MM/DD
Low	Enhancement 6	Other Update: MM/DD

Comments

- Impact to business case
- Impact to value pricing assessment
- Y
- Z

Market Performance: *Project Name*

Business: X

Project #: XYZ

Slide: 16

Summary - Metric				
Product Launch Metrics		Reference Stage	2022 Target	2022 Actuals
Growth	Say / Do Ratio _{Rev} (Revenue \$ Actuals / Forecast)	Launch+	%	%
	Say / Do Ratio _{Mgn} (Margin \$ Actuals / Forecast)		%	%
Profit	Say / Do Ratio _{R&D} (R&D Spend Actuals / Forecast)	Plan	%	%
	Value Capture (List Price - Actual Price) * Quantity	Launch+	\$	\$
Quality	Customer - OTTR	Launch+	%	%
	Mfg Ops - PPM	Launch+	#	#
Comments				
- Growth: X - Profit: X - Quality: X				

Reference Links: Doc 1, Doc 2, Doc 3, Doc 4, Doc 5

Market Optimization		
Product Life Cycle		Estimated Date
Stage	PLC-3: Mature	MM/YYYY
	PLC-4: Decline	MM/YYYY
Performance Risks:		
- Growth: X - Profit: X - Quality: X		
Counter Measures + Next Steps:		
- Growth: X - Profit: X - Quality: X		

Post-Launch R&D Priorities: *Project Name*

Business: X

Project #: XYZ

Slide: 17



Post Launch: VOC

# Customers Contacted	Value Prop Differentiator	Voice of the Customer Learnings	Decision
# ---	XYZ	What did we learn?	Pivot
# ---	XYZ	What did we learn?	Pivot
# ---	XYZ	What did we learn?	Pursue
# ---	XYZ	What did we learn?	Pursue
# ---	XYZ	What did we learn?	Pass
# ---	XYZ	What did we learn?	Pass

Comments

- W
- X
- Y
- Z



Development Priorities

Priority	Feature Enhancement Summary	Timing
High	Enhancement 1	MM/DD
High	Enhancement 2	MM/DD
Med	Enhancement 3	MM/DD
Med	Enhancement 4	MM/DD
Low	Enhancement 5	MM/DD
Low	Enhancement 6	MM/DD

Market Performance - Observations

- W
- X
- Y
- Z

End-of-Life Strategy: *Project Name*

Business: X

Project #: XYZ

Slide: 18

Execution Phases

EOL - 120	EOL - 90	EOL - 60	End of Life
Draft End-of-Life Plan	Align & Execute EOL Plan	Final EOL Deliverables	Communication + Execution
1. Identify 1st Pass EOL/Refresh Plan with draft dates 2. Establish Preliminary Budget Range (plan for upside spend) 3. Host EOL alignment meetings to identify internal & external resourcing needs 4. Schedule follow-up meetings to monitor execution leading to EOL - 90	1. Identify 2nd Pass EOL Plan with aligned dates 2. Establish Approved Budget Range and create new project 3. Host alignment meetings to execute to EOL Plan and create new project as needed. 4. Schedule follow-up meetings to monitor execution leading to EOL - 30, identify Risks.	1. Identify Final Communication Strategy with aligned dates 2. Finalize Approved Budget Range (lower/raise) 3. Continue executing to EOL Plan, escalate as needed. 4. Meet as needed to monitor execution leading to EOL, monitor & report out Risks.	1. Execute Final Communication Strategy to Customers 2. Report out on Budget Actuals vs Plan. Perform AAR. 3. Complete remaining EOL deliverables with Customers. 4. Update EOL and New Project revenue projections, continue to report out Risks.

Reference Links: End-of-Life Plan, Doc 2, Doc 3

04

Approval Templates

Change Summary, PRB Review, E-PAC Executive Approvals

Change Summary: [Project Name](#)

Business: X

Project #: XYZ

Slide: PRB - Changes

3-Year NPV: X.X

Dev Stage: XYZ

Date 1st Revenue: MM/YYYY



Changes Tracking

Gate Review History

Secti	Business Case Highlights	Concieve	Plan	Develop	Qualify	Launch	Maximize	Retire
Market	TAM, \$							
	SAM, \$							
	Market CAGR, %							
Date	Date of 1st Revenue	MM/YY	MM/YY	MM/YY	MM/YY	MM/YY	MM/YY	MM/YY
Financials	3 Year Year 1							
	Incremental Revenues, \$							
	Incremental Margin, \$							
	3 Year NPV, \$							
	Incremental Revenues, \$							
	Incremental Margin, \$							
Value Proposition	3 Year NPV, \$							
	Competitive NBA, \$							
	Competitive NBA Price, \$							
	Top 3 Differentiators	X Y Z	X Y Z	X Y Z	X Y Z	X Y Z	X Y Z	X Y Z
	Value Creation, \$							
	Value Capture, %							
R&D + Risk	Risk: Technical							
	Risk: Commercial							
	R&D Spend,							
R&D Spend + Project	Risk: Technical							
	Risk: Commercial							
	R&D Spend							

Note: Highlight Changes Only in **Red**

PRB Reviews + Approvals: *Project Name*

Business: X

Project #: XYZ

Slide: PRB - Approvals

3-Year NPV: X.X

Dev Stage: XYZ

Date 1st Revenue: MM/YYYY

Prior Gate Review + Approvals

Required	Function	Reviewers Name	Date
●	Product / Business	F. Last	XYZ
●	Finance / FP&A	F. Last	XYZ
●	R&D - Development	F. Last	XYZ
○	Mfg Ops / Supply	F. Last	XYZ
○	Marketing / Sales	F. Last	XYZ
○	Support / Service	F. Last	XYZ
○	Legal / Trade Review	F. Last	XYZ
○	HR Planning	F. Last	XYZ
○	Other	F. Last	XYZ

PRB + Executive Concerns

- X
- Y
- Z

● Required ○ Optional

Reference Links: Doc 1, Doc 2, Doc 3



Current Gate Review + Approvals

Required	Function	Reviewers Name	Date
●	Product / Business	F. Last	XYZ
●	Finance / FP&A	F. Last	XYZ
●	R&D - Development	F. Last	XYZ
●	Mfg Ops / Supply	F. Last	XYZ
○	Marketing / Sales	F. Last	XYZ
○	Support / Service	F. Last	XYZ
○	Legal Review	F. Last	XYZ
○	HR Planning	F. Last	XYZ
○	Other	F. Last	XYZ

PRB + Executive Concerns

- X
- Y
- Z

Product Requirements: *Design Traceability Matrix*



Microsoft Excel
Worksheet

Plan Stage
 • Customer User Stories
 • Customer Requirement Specs

Develop Stage: Design Matrix
 • Team Specification #
 • Design Output #

Dev Team	Design Input #	Specification Input: Target at Launch	Market Input: Stretch Target	Design Output #	Design Output: Definable / Meseurable
Software	CRS-01.IN.SRS.001	Must be able to upload/delete/re-upload video versions to feed customer portal website.	Enable Admin User Log-In/Password to perform upload.	CRS-01.OUT.SRS.001	Upload/Delete 4 videos in < 5 min on 100 MB/s connection.
Software	CRS-02.IN.SRS.002	Doctor office/Lab to provide QR code (encoded format internally) and/or unique KEY to automate patient account creation .	Unique KEY expires between 1-7 days (Admin may change)	CRS-02.OUT.SRS.002	

Qualify Stage: Design Matrix
 • Design Review #
 • Test Run #



Design Review #	Design Review Plan	Verification & Validation Plan	Test Run #	Test Issues	Test Issue Description
DR-2020.08.13-DR.001			TR-2020.08.27-TR.001	TR.001_2020.08.27_IS.001	
DR-2020.08.13-DR.001			TR-2020.08.27-TR.001		



Launch Stage: Design Matrix
 • Design Review #
 • Test Run #

Design Transfer	Design Transfer Confirmation	Design Change	Design Change Description
DT.001_2020.08.24_PRS.001		DC.001_2020.08.30_PRS.001	

Design Matrix allows interaction between development teams while tracking QA, Test Issues, and Changes.